Personal Information

Name	Dr.-Ing. Nils Gumpfer
Date of Birth	May 9, 1994
Place of Birth	Witzenhausen, Germany
Nationality	German
E-Mail	nils.gumpfer@kite.thm.de

Academic and Professional Experience

06/2025 – present	Research Associate (PostDoc), TimeXAI Group, Centre of Competence for Information Technology (KITE), Technische Hochschule Mittelhessen (THM), Friedberg, Germany **Focus Areas:** - Applied explainable AI for time series data (ECG, sensors) - Supervision of doctoral students
05/2019 – 05/2025	Research Associate (Doctorate), Cognitive Information Systems Group, Centre of Competence for Information Technology (KITE), Technische Hochschule Mittelhessen (THM), Friedberg, Germany **Focus Areas:** - Applied Explainable AI for ECG time series - Supervision of students
12/2018 – present	Lecturer, Technische Hochschule Mittelhessen, StudiumPlus, Wetzlar, Germany **Courses taught:** - Machine Learning - Predictive Analytics - Operating Systems - Fundamentals of Computer Science
06/2018 – 04/2019	Inhouse Consultant for Business Analytics, B. Braun Melsungen AG, Melsungen, Germany **Focus Areas:** - Business Analytics - Project Management - Machine Learning - IT Service Management
07/2016 – 05/2018	Working Student, B. Braun Melsungen AG, Melsungen, Germany **Focus Areas:** - Business Analytics - Project Management - IT Service Management

Education

05/2019 – 06/2025	Doctor of Engineering (Dr.-Ing.), Graduate Centre for Engineering Sciences, Research Campus of Central Hessen, Giessen. Lead institution: Philipps-University of Marburg **Grade:** 0.7 (summa cum laude) **Thesis:** *Explainable Artificial Intelligence for Detection of Structural Changes in Myocardium*
10/2016 – 01/2019	M.Sc. in Business Informatics, Technische Hochschule Mittelhessen, Friedberg **Grade:** 1.0 (very good) **Thesis:** *Prediction of Myocardial Tissue Condition using 12-Lead ECG and Deep Learning*
08/2013 – 06/2016	B.A. in Business Administration (dual program), specialization in Business Informatics, THM StudiumPlus, Bad Wildungen; B. Braun Melsungen AG, Melsungen **Grade:** 1.2 (very good) **Thesis:** *Future Scenarios for Analytical Applications at B. Braun Melsungen AG*
08/2010 – 06/2013	Technical Secondary School (IT track), Max-Eyth-Schule, Kassel **Grade:** 1.1 (very good) **Graduation:** Abitur (A-levels) Certificate **Special Project:** *Development of an Autonomous Chain Robot*
08/2004 – 07/2010	Comprehensive School, Valentin-Traudt-Schule, Grossalmerode **Grade:** 1.0 (very good) **Graduation:** Secondary School Certificate

Awards and Honors

05/2026	Dissertation Price, German Informatics Society: Finalist
01/2025	Erasmus+ Mobility Scholarship (IIIT Allahabad, India)
02/2023	Idea Competition, Philipps-University of Marburg: 2nd Prize
11/2021	HessenIdeen Start-Up Competition: 2nd prize
10/2021	Idea Slam, Justus-Liebig- University of Giessen: 2nd Prize
09/2022	Hessian Founder's Award: Semi-Finalist
09/2021	Startup Competition, THM: 1st Prize
05/2021-11/2021	HessenIdeen Start-Up Scholarship
07/2019	Best Thesis Award (M. Sc. Business Informatics), THM

Research Projects

09/2024-09/2027	*TimeXAI* – Explainable AI for Time Series **Focus:** - Applied explainable AI for time series data (ECG, sensors) - Groundwork - User studies **Funding:** German Federal Ministry of Research, Technology and Space
10/2024-10/2025	*SUSTAIN-AI-ECG / ECG4Africa.org* – ECG AI Application **Focus:** - Deployment of ready-to-use ECG application - DevOps, system integration - AI-based annotations, ECG viewer - User-studies with clinical personnel **Funding:** Bavarian State Chancellery
05/2022-04/2024	*RisKa* – Risk Stratification in Cardiology **Focus:** - Development of AI models for ECG analysis - XAI for user-trust - Startup founding preparation **Funding:** Hessian Minsitry of Digiatlization and Innovation
10/2022-09/2025	*HERMIQS* – Heart Emergency Rescue Management **Focus:** - Development of ready-to-use ECG application - DevOps, system integration - User-studies with emergency personnel **Funding:** Hessian Minsitry of Digiatlization and Innovation

Publications

Journal Articles (peer-reviewed)

- Gumpfer N., Grün D., Hannig J., Keller T., and Guckert M. (2020c). “Detecting Myocardial Scar Using Electrocardiogram Data and Deep Neural Networks.” *Biological Chemistry* 402.8, pp. 911–923. DOI: 10.1515/hsz-2020-0169.
- Guckert M., Gumpfer N., Hannig J., Keller T., and Urquhart N. (2021). “A Conceptual Framework for Establishing Trust in Real World Intelligent Systems.” *Cognitive Systems Research* 68, pp. 143–155. DOI: 10.1016/j.cogsys.2021.04.001.
- Gruen D., Rudolph F., Gumpfer N., Hannig J., Elsner L. K., von Jeinsen B., Hamm C. W., Rieth A., Guckert M., and Keller T. (2021). “Identifying Heart Failure in ECG Data With Artificial Intelligence - A Meta-Analysis.” *Frontiers in Digital Health* 2, p. 67. DOI: 10.3389/fdgth.2020.584555.
- Gumpfer N., Prim J., Keller T., Seeger B., Guckert M., and Hannig J. (2023). “SIGNed Explanations: Unveiling Relevant Features by Reducing Bias.” *Information Fusion* 99, p. 101883. DOI: 10.1016/j.inffus.2023.101883.
- Powers S. T., Linnyk O., Guckert M., Hannig J., Pitt J., Urquhart N., Ekárt A., Gumpfer N., Han T. A., Lewis P. R., Marsh S., and Weber T. (2023). “The Stuff We Swim in: Regulation Alone Will Not Lead to Justifiable Trust in AI.” *IEEE Technology and Society Magazine* 42.4, pp. 95–106. DOI: 10.1109/MTS.2023.3341463.

Conference Papers (peer-reviewed)

- Gumpfer N., Prim J., Gruen D., Hannig J., Keller T., and Guckert M. (2021d). “An Experiment Environment for Definition, Training and Evaluation of Electrocardiogram-Based AI Models.” In: *19th International Conference on Artificial Intelligence in Medicine, AIME 2021, Virtual Event, June 15 - 18, 2021, Proceedings*. Ed. by A. Tucker, P. H. Abreu, and J. Cardoso. Vol. 12721. Lecture Notes in Computer Science. Springer Nature Switzerland AG. Chap. 45, pp. 384–388. DOI: 10.1007/978-3-030-77211-6_45.
- Uhlemann T., Prim J., Gumpfer N., Grün D., Wegener S., Krug S., Hannig J., Keller T., and Guckert M. (2021). “An Ensemble Learning Approach to Detect Cardiac Abnormalities in ECG Data Irrespective of Lead Availability.” In: *Computing in Cardiology (CinC) 2021, Brno, Czech Republic, September 13-15, 2021*. IEEE, pp. 1–4. DOI: 10.23919/CinC53138.2021.9662807.
- Prim J., Uhlemann T., Gumpfer N., Grün D., Wegener S., Krug S., Hannig J., Keller T., and Guckert M. (2021c). “A Data Pipeline for Extraction and Processing of Electrocardiogram Recordings.” In: *Computing in Cardiology (CinC) 2021, Brno, Czech Republic, September 13-15, 2021*. IEEE, pp. 1–4. DOI: 10.23919/CinC53138.2021.9662946.
- Gumpfer N., Dinov B., Sossalla S., Guckert M., and Hannig J. (2024a). “Towards Trustworthy AI in Cardiology: A Comparative Analysis of Explainable AI Methods for Electrocardiogram Interpretation.” In: *22nd International Conference on Artificial Intelligence in Medicine, AIME 2024, Salt Lake City, UT, USA, July 9 - 12, 2024, Proceedings*. Ed. by J. Finkelstein, R. Moskovitch, and E. Parimbelli. Vol. 14845. Lecture Notes in Computer Science. Springer Nature Switzerland AG. Chap. 36, pp. 350–361. DOI: 10.1007/978-3-031-66535-6_36.

Posters

- Gumpfer N, Prim J., Hannig J., Grün D., Morgen A., Körber M., B.Seeger, Keller T., and Guckert M. (2020b). *Using database queries to detect conspicuous R-R-intervals in electrocardiogram records*. Poster contribution, 36th German Conference on Bioinformatics, Frankfurt (Main), Germany, September 14 - 17, 2020. Poster P52.
- Hannig J., Gumpfer N., Prim J., Grün D., Keller T., and Guckert M. (2020). *A deep learning approach to predict myocardial scar tissue based on electrocardiogram data*. Poster contribution, 36th German Conference on Bioinformatics, Frankfurt (Main), Germany, September 14 - 17, 2020. Poster P26.
- Gumpfer N, Wegener S, Prim J, Gruen D, Hannig J, Keller T, and Guckert M (2021b). *On the importance of representative datasets in ECG-based artificial intelligence*. Poster contribution, ESC Congress 2021 - The Digital Experience, Virtual Event, August 27 - 30, 2021. DOI: 10.1093/eurheartj/ehab724.3060.
- Prim J, Uhlemann T, Gumpfer N, Gruen D, Wegener S, Krug S, Hannig J, Keller T, and Guckert M (2021a). *A data-pipeline processing electrocardiogram recordings for use in artificial intelligence algorithms*. Poster contribution, ESC Congress 2021 - The Digital Experience, Virtual Event, August 27 - 30, 2021. DOI: 10.1093/eurheartj/ehab724.3041.
- Gumpfer N, Prim J., Gruen D., Hannig J., Keller T., and Guckert M. (2021a). *An Experiment Environment for Definition, Training and Evaluation of Electrocardiogram-Based AI Models*. Poster contribution, 19th International Conference on Artificial Intelligence in Medicine, Virtual Event, June 15 - 18, 2021. DOI: 10.1007/978-3-030-77211-6_45.
- Wegener S, Gruen D, Prim J, Gumpfer N, Wolter J. S., Hamm C. W., Liebetrau C, Hannig J, Guckert M, and Keller T (2021a). *Predicting mortality in cardiovascular patients using electrocardiogram data and artificial intelligence*. Poster contribution, ESC Congress 2021 - The Digital Experience, Virtual Event, August 27 - 30, 2021. DOI: 10.1093/eurheartj/ehab724.1132.
- Wegener S, Schmidt T, Prim J, Gumpfer N, Gruen D, Hannig J, Guckert M, and Keller T (2021c). *Detecting a broader spectrum of cardiac pathologies in electrocardiogram data by applying a deep neural network designed to detect a specific cardiac disease*. Poster contribution, ESC Congress 2021 - The Digital Experience, Virtual Event, August 27 - 30, 2021. DOI: 10.1093/eurheartj/ehab724.3063.
- Wegener S, Grün D, Gumpfer N, Prim J, Hannig J, Guckert M, and T Keller und (2022). *KardioIQ: Eine interdisziplinäre hochschulübergreifende Forschergruppe zum Thema Künstliche Intelligenz in der Kardiologie*. Poster contribution, Justus-Liebig-University Gießen Science Day 2022, Gießen, Germany, November 11, 2022.
- Dittrich M, Hannig J, Grün D, Wegener S, Gumpfer N, Stützner E, Prim J, Guckert M, and Keller T (2022). *Estimation of the Individual Heart Age using a on 12-lead ECG Data Trained Neural Network*. Poster contribution, Justus-Liebig-University Gießen Science Day 2022, Gießen, Germany, November 11, 2022.

Conference Abstracts

- Gumpfer N, Hurka S., Hannig J., Rolf A., Hamm C. W., Guckert M., and Keller T. (2019b). "Development of a machine learning algorithm to predict myocardial scar based on a 12-lead electrocardiogram." In: *85th Annual Meeting of the German Cardiac Society - Cardiac and Circulation Research, Mannheim, Germany, April 24 - 27, 2019*. Vol. 108. Clinical Research in Cardiology Supplement 1. Abstract V1802. German Cardiac Society (DGK). Springer. DOI: 10.1007/s00392-019-1435-9.

- Grün D., Rudolph F., Gumpfer N., Hannig J., Elsner L. K., Jeinsen B. von, Rieth A., Hamm C. W., Guckert M., and Keller T. (2020). "Use of artificial intelligence to process ECG data for identification of heart failure - a meta-analysis." In: *86th Annual Meeting of the German Cardiac Society - Cardiac and Circulation Research, Berlin, Germany, October 14 - 17, 2020*. Vol. 109. Clinical Research in Cardiology Supplement 1. Abstract V633. German Cardiac Society (DGK). Springer. DOI: 10.1007/s00392-020-01621-0.
- Gumpfer N, Wegener S, Prim J, Gruen D, Hannig J, Keller T, and Guckert M (2021c). "On the importance of representative datasets in ECG-based artificial intelligence." In: *ESC Congress 2021 - The Digital Experience, August 27 - 30, 2021, Abstract Supplement*. Vol. 42. European Heart Journal Supplement 1. Abstract ehab724.3060. European Society of Cardiology (ESC). Oxford University Press. DOI: 10.1093/eurheartj/ehab724.3060.
- Wegener S, Schmidt T, Prim J, Gumpfer N, Gruen D, Hannig J, Guckert M, and Keller T (2021d). "Detecting a broader spectrum of cardiac pathologies in electrocardiogram data by applying a deep neural network designed to detect a specific cardiac disease." In: *ESC Congress 2021 - The Digital Experience, August 27 - 30, 2021, Abstract Supplement*. Vol. 42. European Heart Journal Supplement 1. Abstract ehab724.3063. European Society of Cardiology (ESC). Oxford University Press. DOI: 10.1093/eurheartj/ehab724.3063.
- Prim J, Uhlemann T, Gumpfer N, Gruen D, Wegener S, Krug S, Hannig J, Keller T, and Guckert M (2021b). "A data-pipeline processing electrocardiogram recordings for use in artificial intelligence algorithms." In: *ESC Congress 2021 - The Digital Experience, August 27 - 30, 2021, Abstract Supplement*. Vol. 42. European Heart Journal Supplement 1. Abstract ehab724.3041. European Society of Cardiology (ESC). Oxford University Press. DOI: 10.1093/eurheartj/ehab724.3041.
- Wegener S, Gruen D, Prim J, Gumpfer N, Wolter J. S., Hamm C. W., Liebetrau C, Hannig J, Guckert M, and Keller T (2021b). "Predicting mortality in cardiovascular patients using electrocardiogram data and artificial intelligence." In: *ESC Congress 2021 - The Digital Experience, August 27 - 30, 2021, Abstract Supplement*. Vol. 42. European Heart Journal Supplement 1. Abstract ehab724.1132. European Society of Cardiology (ESC). Oxford University Press. DOI: 10.1093/eurheartj/ehab724.1132.
- Stützner E., Gumpfer N., Wegener S., Prim J., Grün D., Hanning J., Guckert M., and Keller T. (2022). "Machine learning to optimize ECG leadwise evaluation for localization of myocardial infarction." In: *88th Annual Meeting of the German Cardiac Society - Cardiac and Circulation Research, Mannheim, Germany, April 20 - 23, 2022*. Vol. 111. Clinical Research in Cardiology Supplement 1. Abstract V982. German Cardiac Society (DGK). Springer. DOI: 10.1007/s00392-022-02002-5.

Oral Presentations and Invited Talks

- Gumpfer N. (2025b). *From Saliency to Semantics: XAI for ECG Time Series Analysis*. Invited talk, IEEE SPS Cycle 2 Seasonal School on Explainable AI and Applications to Biometric Signal Processing, IIIT Allahabad, Prayagraj, India, July 17, 2025.
- Gumpfer N. (2025c). *KI-gestützte EKG-Analyse*. Invited talk, KI made in Hessen, ZukunftsRaum Friedberg, Friedberg, Hesse, Germany, June 17, 2025.
- Gumpfer N. (2025a). *Balancing Explanations with SIGN*. Oral presentation, 2nd Colloquium on Business Informatics, Indo-German Workshop on XAI & Federated Learning, IIIT Allahabad, Prayagraj, India, January 21–23, 2025.
- Gumpfer N. (2024a). *Erklärbare Künstliche Intelligenz*. Invited talk, Digitalkonferenz des Wetteraukreises 2024, Ortenberg, Hesse, Germany, November 27, 2024.

- Gumpfer N., Dinov B., Sossalla S., Guckert M., and Hannig J. (2024b). *Towards Trustworthy AI in Cardiology: A Comparative Analysis of Explainable AI Methods for Electrocardiogram Interpretation*. Oral presentation, 22nd International Conference on Artificial Intelligence in Medicine, AIME 2024, Salt Lake City, UT, USA, July 9 - 12, 2024.
- Gumpfer N. (2024c). *Opening the AI Black Box to Build Trust in Cardiology*. Poster presentation, Get-together of the Graduate Centre for Engineering Sciences at the Research Campus of Central Hessen, April 24, 2024, Gießen, Germany.
- Gumpfer N. (2021). *Erklärbare Künstliche Intelligenz zur Erkennung von Herzmuskelschädigungen*. Oral presentation, 9th Interdisciplinary Doctoral Colloquium at Technische Hochschule Mittelhessen, Gießen, Germany, October 28, 2021.
- Gumpfer N., Wegener S., Grün D., Stützner E., Prim J., Hannig J., Guckert M., and Keller T. (2022). *Explainable artificial intelligence for detection of STEMI*. Oral presentation, Justus-Liebig-University Gießen Science Day 2022, Gießen, Germany, November 11, 2022.
- Gumpfer N, Gruen D., Hannig J., Keller T., and Guckert M. (2020a). *Detecting Myocardial Scar Using Electrocardiogram Data and Deep Neural Networks*. Oral presentation, 36th German Conference on Bioinformatics, Frankfurt (Main), Germany, September 14 - 17, 2020.

Doctoral Thesis

- Gumpfer N. (2024b). *Explainable Artificial Intelligence for Detection of Structural Changes in Myocardium*. Ed. by B. Seeger and M. Guckert. Doctoral Thesis. Philipps-Universität Marburg. DOI: 10.17192/z2025.0472.

Open Source Software

- Experiment environment proposed in paper *"An Experiment Environment for Definition, Training and Evaluation of Electrocardiogram-Based AI Models"* (Gumpfer et al., 2021d), available at:
  https://github.com/nilsgumpfer/experiment-environment-ecg-ai
- Experiment code related to paper *"SIGNed explanations: Unveiling relevant features by reducing bias"* (Gumpfer et al., 2023), available at:
  https://github.com/nilsgumpfer/SIGN
- Python package including all XAI methods used in paper *"SIGNed explanations: Unveiling relevant features by reducing bias"* (Gumpfer et al., 2023), available at:
  https://pypi.org/project/signxai/
  https://github.com/nilsgumpfer/SIGN-XAI
- Python package with SIGN variants for PyTorch, available at:
  https://pypi.org/project/signxai2/
  https://github.com/TimeXAIgroup/signxai2
- Experiment code related to paper *"Towards Trustworthy AI in Cardiology: A Comparative Analysis of Explainable AI Methods for Electrocardiogram Interpretation"* (Gumpfer et al., 2024a), available at:
  https://github.com/nilsgumpfer/AIME2024

- Experiment code related to IEEE tutorial *"From Saliency to Semantics: XAI for ECG Time Series Analysis"* (Gumpfer, 2025b), available at:
  https://github.com/nilsgumpfer/ieee_sps_xai_2025
- Experiment code related to lecture *"Machine Learning"*, available at:
  https://github.com/nilsgumpfer/machine_learning_stplus
- Experiment code related to lecture *"Predictive Analytics"*, available at:
  https://github.com/nilsgumpfer/predictive_analytics_stplus
- Demo code related to lecture *"Operating Systems"*, available at:
  https://gitlab.com/nilsgumpfer/esp8266-chat
- Demo code for science communication project on *"XAI for Time Series"*, available at:
  https://gitlab.com/nilsgumpfer/pendulum-timeseries
- Demo code for science communication project on *"Computer Vision with YOLO"*, available at:
  https://gitlab.com/nilsgumpfer/yolo-demo